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Deep Convolutional Neural Network (DCNN) Models for Accurate Analysis and Labeling of Complex Reservoir Temporal Data Sequences

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Abstract

Reservoir engineering relies heavily on understanding patterns and signatures in temporal data to inform field development and management decisions. However, classical data analysis and mining methods often struggle to extract meaningful insights from complex, high-dimensional data. This paper proposes a novel approach using Deep Convolutional Neural Networks (DCNNs) to analyze and label temporal reservoir data, enabling more accurate and efficient decision-making.

DCNNs are used in many applications such as image classification, facial recognitions, and medical image analysis. In this work, DCNNs are extended to applications in oil and gas to recognize patterns, identify signatures, classify, and label the reservoir temporal data. Temporal reservoir data (e.g., production/injection data, pressures, water-cut and gas-oil ratios) exhibits complex, nonlinear patterns and variable signatures, making it challenging to analyze. DCNN models were developed to work with distinct vector types (e.g., Water and Oil Production Rates and Water-Cut) to predict labels specific to each vector type. For example, in the case of the Oil Production Rates vector, the model predicts labels such as ‘Under producing’, ‘Late matching’, or ‘Matching’. The development of the DCNN models involved utilizing hundreds of images of temporal data to train, validate and test the neural networks. Those images are in the form of line plot images, which were previously labeled. This process also involves data preprocessing, feature extraction, and training using a combination of convolutional and fully connected layers to learn complex patterns in the data.

This work utilized 80 realizations of NORNE and BRUGGE benchmark models to demonstrate the efficiency of Deep Convolutional Neural Network (DCNN) models in analyzing reservoir temporal data. Experiments with different numbers of images showed the robustness of the models, with accuracy improving from 89%, using 3900 images, to 97% using 250,000 images. This also demonstrates the models’ ability to learn complex data patterns with high accuracy. The approach enabled the DCNN models to perform independently, without relying on external labeling processes. This has significantly reduced the time, effort, and resources required by engineers, as compared to classical data analysis and mining methods. Additionally, the models are field-independent, allowing them to be applied across different reservoir models and fields. This enables DCNN to evaluate wells and field performance during processes involving reservoir models’ calibration and history matching.

This study presents a novel application of Deep Convolutional Neural Networks (DCNNs) in reservoir engineering, leveraging their pattern recognition capabilities to analyze temporal data. By extending DCNNs to oil and gas applications, this research provides a new framework for an accurate temporal data labeling, enabling optimized model calibration and history matching processes, as well as evaluating field development scenarios.

Introduction

The effective management and development of oil and gas fields depend on the ability to accurately extract valuable insights from complex temporal data. Temporal data, which includes production rates, pressure, water cut, and gas-oil ratios, is critical for understanding reservoir behavior, optimizing production, and predicting future performance. However, analyzing temporal data is a complex task, as it involves dealing with large amounts of data, non-linear relationships, and multiple variables. Conventional methods of analyzing temporal data, such as statistical analysis and data mining, have been widely used but often struggle to handle complex, high-dimensional data and require significant manual interpretation. To address this limitation, there is a need for more innovative and automated methods to analyze and interpret temporal data, providing more accurate and reliable insights for decision-making and ultimately leading to improved field development and management outcomes.

In recent years, artificial intelligence (AI) has revolutionized numerous industries by enabling machines to learn from complex data, recognize patterns, and make accurate predictions with minimal human intervention. Image recognition and classification have been extensively studied in various fields, including computer vision, medical imaging, and remote sensing. Traditional methods for image recognition and classification include hand-crafted feature extraction techniques, such as edge detection, texture analysis, and shape description, followed by classification using machine learning algorithms like support vector machines (SVMs) and k-nearest neighbors (KNNs). However, these traditional methods have limitations, such as requiring manual feature engineering and being sensitive to variations in lighting, pose, and other environmental factors.

Deep learning (DL) techniques have shown to be more efficient and accurate than traditional methods for image recognition and classification. Convolutional neural networks (CNNs) have become a popular choice for image classification tasks, due to their ability to automatically learn features from raw images. CNNs have been successfully applied to various image classification tasks, including object recognition, facial recognition, and medical image analysis (Ayeni, J. 2022). In the medical field, DL has also been used to automatically detect and quantify blood vessels tortuosity (da Silva et al. 2022). Han (2021) demonstrated a DL-based approach to detect brain tumors from magnetic resonance imaging (MRI) scans.

Building on these advancements, AI and DL have also made significant contributions to reservoir engineering. Traditional modeling and forecasting methods, which rely on physics-based simulations and statistical techniques, often struggle with the complexity and noise inherent in reservoir data. The application of Artificial Intelligence in the oil and gas industry has gained significant attention, with successful applications in exploration, drilling, production, and reservoir management (Alkinani et al., 2019). ANNs have been used to analyze complex data and make predictions, such as predicting rate of penetration (Al-AbdulJabbar et al., 2018) and estimating reservoir properties (Kohli & Arora, 2014). Furthermore, Ghanduri (2024) utilized Dynamic Time Warping (DTW) with K-Nearest Neighbor (KNN) to classify ESP gas lock, demonstrating DTW's effectiveness in measuring time series similarity which achieved 87% accuracy, showcasing DTW's potential in improving time series classification.

Deep learning techniques have shown to be more efficient and accurate than traditional reservoir numerical simulation methods for predicting reservoir dynamic performance. According to Wang et al. (2023), deep learning models can predict remaining oil saturation distribution with high accuracy and faster than reservoir numerical simulation. Additionally, machine learning and deep learning approaches

have been explored for forecasting oil production and bottom-hole pressure (BHP) in hydrocarbon reservoirs, providing an alternative to traditional reservoir simulation models that are time-consuming and computationally expensive (Werneck et al., 2019).

The use of deep learning techniques in reservoir modeling and simulation has several advantages, including the ability to handle large datasets, capture complex relationships, and provide accurate forecasts. However, there are also challenges associated with the application of deep learning techniques, such as the need for large amounts of training data, the risk of overfitting, and the need for careful model selection and hyperparameter tuning. This paper aims to investigate the potential of deep convolutional neural networks (DCNNs) in recognizing patterns and classifying temporal reservoir data, enabling more accurate and efficient decision-making in reservoir engineering.

Methodology

Data Collection

The dataset used in this study consisted of a large collection of temporal reservoir data for distinct vectors, collected from 80 realizations of NORNE and BRUGGE. The dataset consisted of a total of 255,780 images of wells' Water-cut vector and 262,965 images of wells' Oil production rate.

Data Preprocessing

Data preprocessing is a crucial step in improving the efficiency of any classification problem. To prepare the temporal reservoir data for training a Deep Convolutional Neural Network (DCNN) model, the data underwent rigorous preprocessing. The data was extracted and normalized to ensure a consistent scale between 0 and 1, preventing features with large ranges from dominating the model. This step is crucial for maintaining the integrity and balance of the dataset. The Min-Max Scaler normalization method was employed, using the formula:

$$x' = (x - \min) / (\max - \min)$$

where x' is the normalized value, x is the original value, $\min$ is the minimum value in the dataset, and $\max$ is the maximum value in the dataset.

The preprocessed data was then resized to (64×64) pixels, to have a good balance between preserving detailed information and reducing computational requirements. The WWCT dataset was labeled with categories including "High Water-cut", "Low Water-cut", "Early Water Breakthrough", "Late Water Breakthrough", and "Matching". The WOPR dataset was labeled with categories including "Over Producing", "Under Producing", and "Matching". To facilitate model training, label encoding was performed to map these categorical labels into numerical values for both datasets. And finally, the dataset was divided into training and testing datasets, where 80% of the dataset was for training the model and 20% for testing. Fig.1 shows an example of the dataset for multi-classification.

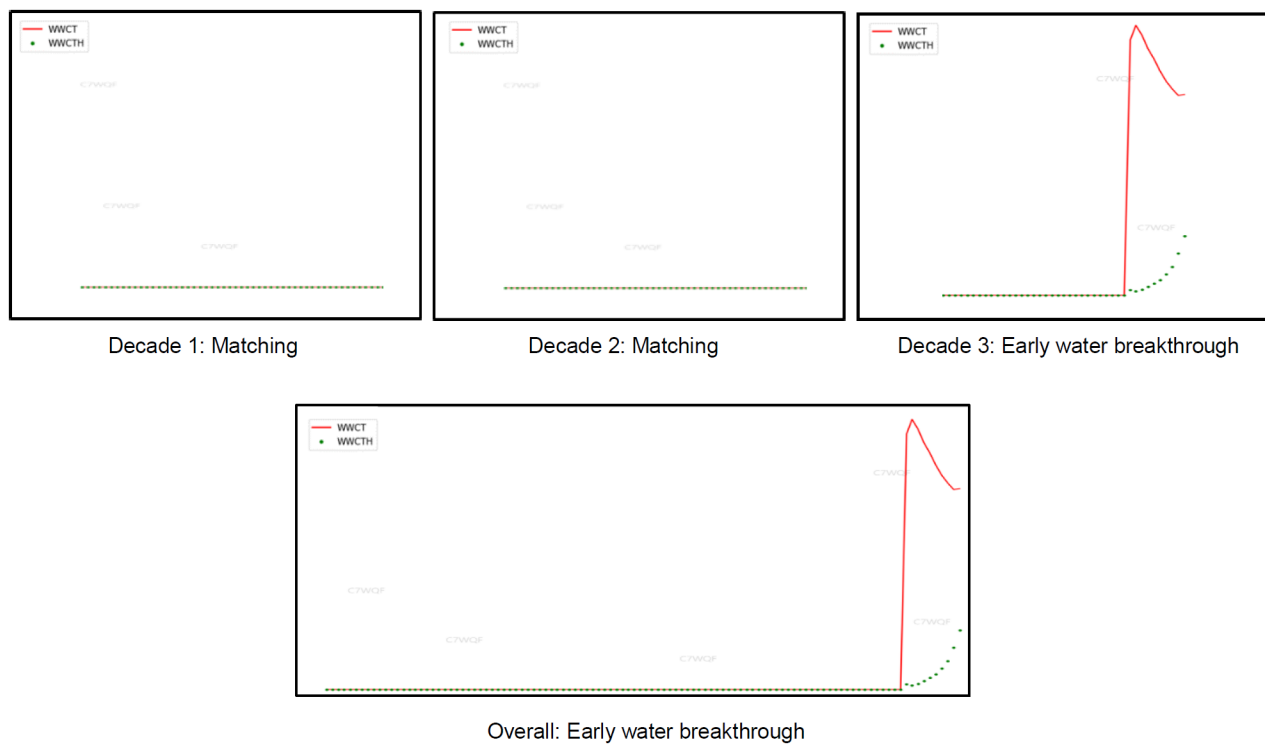


Figure 1—Multi-classification example for Water-cut.

Fig.2 shows an example of binary classification of water-cut.

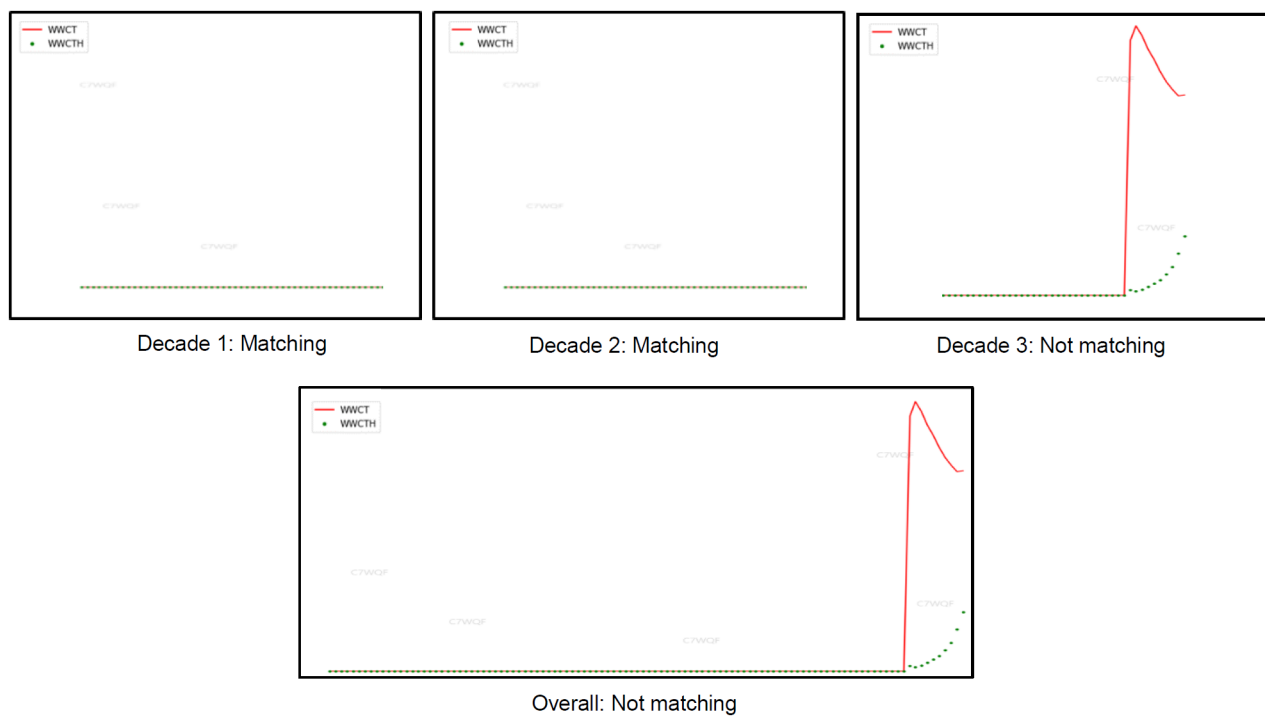


Figure 2—Binary classification example for Water-cut.

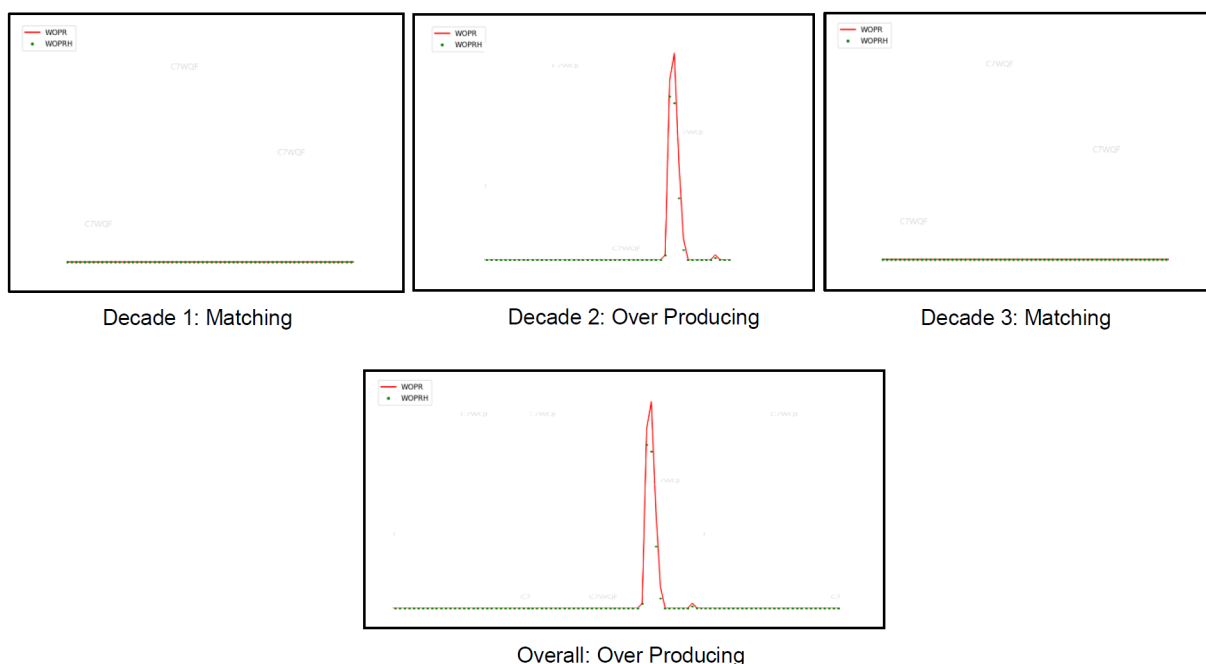


Figure 3—shows an example of multi-classification of Well-Oil production rate.

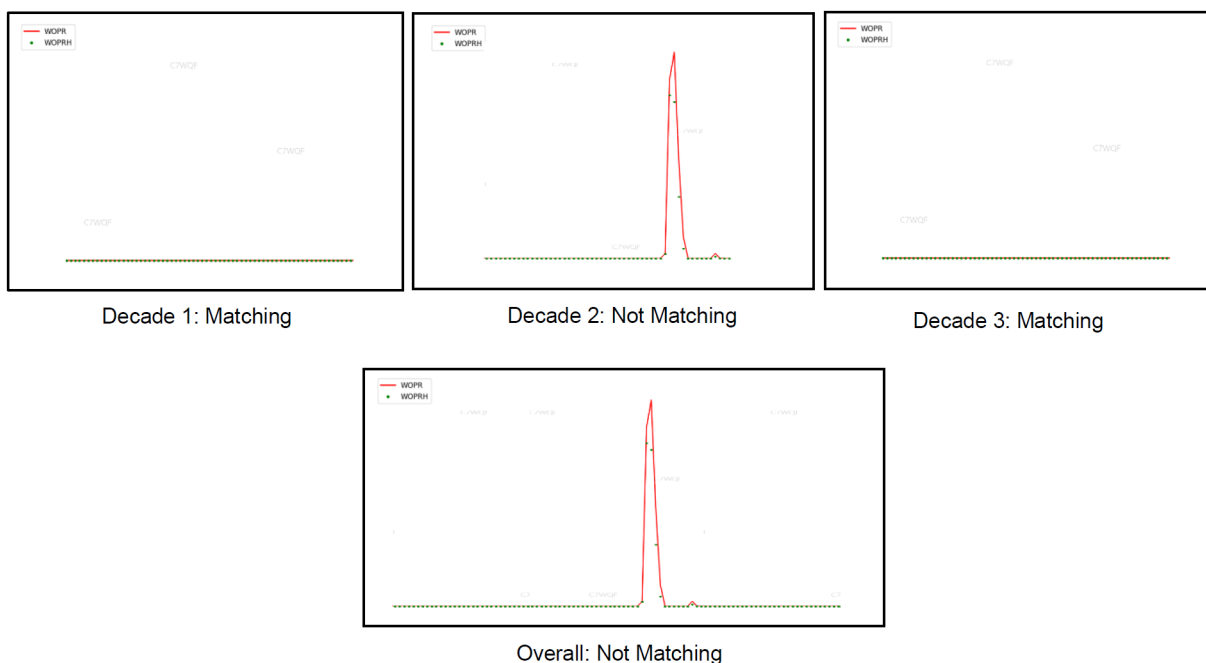


Figure 4—Binary classification of Well-Oil production rate.

DCNN Architecture

A Deep Convolutional Neural Network (DCNN) is a type of neural network that uses convolutional and pooling layers to analyze data with grid-like topology, such as images. DCNNs work by applying a series of convolutional and pooling layers to the input data, followed by fully connected layers to produce the output. The convolutional layers apply a set of filters to the input data, scanning the data in a sliding window fashion, to generate feature maps. The pooling layers downsample the feature maps to reduce spatial dimensions and retain important information. The fully connected layers then classify the output based on the features extracted by the convolutional and pooling layers (Ankile et al. 2020). In this study, a DCNN was utilized for the classification of temporal reservoir data, with the model's performance being heavily dependent

on the size of the training dataset. A dataset comprising over 255,780 images was found to be sufficient for adequately training the DCNN model. The DCNN architecture employed in this study consisted of a sequence of layers, including Convolutional, Pooling, Flatten, Dropout, and Dense, which are briefly described below (Goodfellow et al, 2016):

- **Convolutional Layer:** The convolutional layer is a fundamental component of a Convolutional Neural Network. It applies a set of learnable filters (kernels) to the input data, which can be an image or a feature map. Each filter produces a feature map by sliding over the input spatially and computing weighted sums. This process is known as convolution which allows the convolutional layer to capture features from multiple channels, such as color images with RGB channels. The resulting feature map represents the presence of specific features, such as edges or lines, at different locations in the input data.
- **Pooling Layer:** The pooling layer is used to reduce the spatial dimensions of the feature maps generated by the convolutional layer. This is done to retain the most important information and discard the less relevant details. The most common type of pooling is max pooling, which involves dividing the feature map into smaller regions and selecting the maximum value from each region.
- **Flatten Layer:** The flatten operation is an essential step in the neural network pipeline, involving the reshaping of the multidimensional output of the convolutional and pooling layers into a one-dimensional array, which is necessary to be fed into the fully connected layers that require a one-dimensional input, allowing them to process the data and make predictions.
- **Dropout Layer:** The dropout layer is a regularization technique used to prevent overfitting in neural networks. Overfitting occurs when the network becomes too specialized to the training data and fails to generalize well to new, unseen data. The dropout layer helps to prevent this by randomly dropping out a fraction of the neurons during training.
- **Fully Connected (Dense) Layer:** The fully connected layer, also known as the dense layer, is last layer of the neural network architecture where each neuron receives input from all outputs of the previous layers. This layer is responsible for making predictions based on the features extracted by the convolutional and pooling layers. Fig.5 shows a general flow between all the layers of CNN.

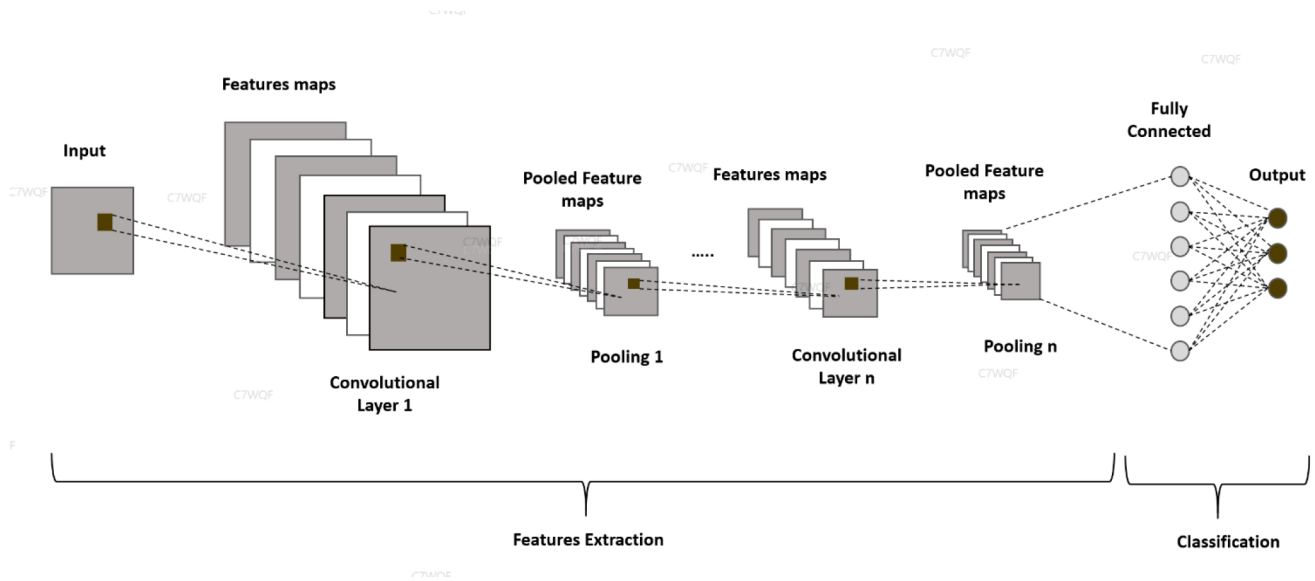


Figure 5—Figure shows a general flow between all the layers of CNN.

DCNN Model Development

The DCNN architecture was developed to train a model that can classify trends of the temporal reservoir data. The architecture consists of a convolutional layer with a 'relu' activation function to introduce non-linearity into the model, which allows it to learn more complex relationships between inputs and outputs. The model also features multiple of convolutional layers with Pooling and Dropout layers to reduce overfitting and improve generalization. Finally, the output of the convolutional layers is then flattened and passed through the fully connected layer, which uses a "softmax" activation function for multi-class classification and a "sigmoid" function for binary classification.

DCNN Model Training

The purpose of the model training procedure is to discover the best set of parameters for generalizing new data while avoiding overfitting and underfitting. Several optimization approaches were used to train the model during this phase. The "categorical_crossentropy" and "binary_crossentropy" loss functions were used to enable the optimization algorithms, such as Adam, to adjust the model parameters during training and generate the least potential loss. Furthermore, an appropriate learning rate was set to assist Adam in updating the model parameters. To prevent overfitting, we employed a technique called early stopping, which stops the training process when the model's performance on the validation set starts to degrade. This technique helps prevent the model from overfitting to the training data and improves its generalization to new, unseen data. By using early stopping, we can avoid overfitting and ensure that our model is able to make accurate predictions on new data.

Transfer Learning Models

Transfer learning is a machine learning technique that allows leveraging the knowledge gained from pre-trained models to improve the performance of models. This is achieved by using the pre-trained model as a starting point and fine-tuning it to fit a specific task. The pre-trained model has already learned to recognize certain features and patterns from the large dataset it was trained on, and by using this as a foundation, it is possible to adapt it to a new dataset and task (Houda Bichria et al, 2023). In this study, we employed two widely-used CNN architectures, **MobileNetV2** and **VGG16**, to classify temporal reservoir data, demonstrating the effectiveness of transfer learning in image classification tasks.

- **MobileNetV2 Model:** As demonstrated by (Houda Bichria et al, 2023), MobileNetV2 is a convolutional neural network architecture that is designed to be efficient and effective for image classification tasks. It uses inverted residual blocks with bottlenecking features, which allows it to reduce the number of parameters and computations required while maintaining high accuracy. This architecture is particularly useful for mobile and embedded vision applications, where computational resources are limited. MobileNetV2 has been shown to achieve high accuracy in image classification tasks, making it a popular choice for a wide range of applications.
- **VGG16 Model:** VGG16 is a convolutional neural network comprising 16 layers, which uses multiple 3×3 filters in each convolutional layer. This architecture makes VGG16 a popular method for image classification due to its ability to extract features effectively. VGG16 is trained on the ImageNet database, which contains a million images of 1000 categories, and has been widely used as a pre-trained model for various image classification tasks.

Results

Model Evaluation Metrics

The performance of deep-learning models is typically evaluated using metrics such as accuracy and error, which provide insight into the relationship between predicted and actual values. Accuracy is a key

performance indicator that is calculated by dividing the number of correctly classified instances, comprising true positives and true negatives, by the total number of classified instances. This can be expressed mathematically as:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

where true positives represent instances where both actual and predicted values are positive, true negatives represent instances where both actual and predicted values are negative, false positives occur when the actual value is negative but the predicted value is positive, and false negatives occur when the actual value is positive but the predicted value is negative.

DCNN Model Evaluation

The results of the developed DCNN model are presented in Table 1, which demonstrates the model's performance in classifying temporal reservoir data. The model achieved an accuracy of 97% for the WWCT vector and 98% for the WOPR vector in the multi-classification scenario. In the binary classification scenario, the DCNN model exhibited a high degree of accuracy, achieving 98% for both the WWCT and WOPR vectors. These results indicate the model's effectiveness in accurately classifying temporal reservoir data.

Table 1—The results of DCNN model.

Model	Multi classification	Binary classification
DCNN for WWCT	97%	98%
DCNN for WOPR	98%	98%

Fig.6 shows the training history of the DCNN model for WWCT multi-classification, showcasing the accuracy and loss plots for the training and validation datasets across the training epochs, respectively. It displays that the DCNN model starts to converge after 46 epochs for both training and validation phases, while the loss begins to decrease, approaching zero, as the model optimizes its performance.

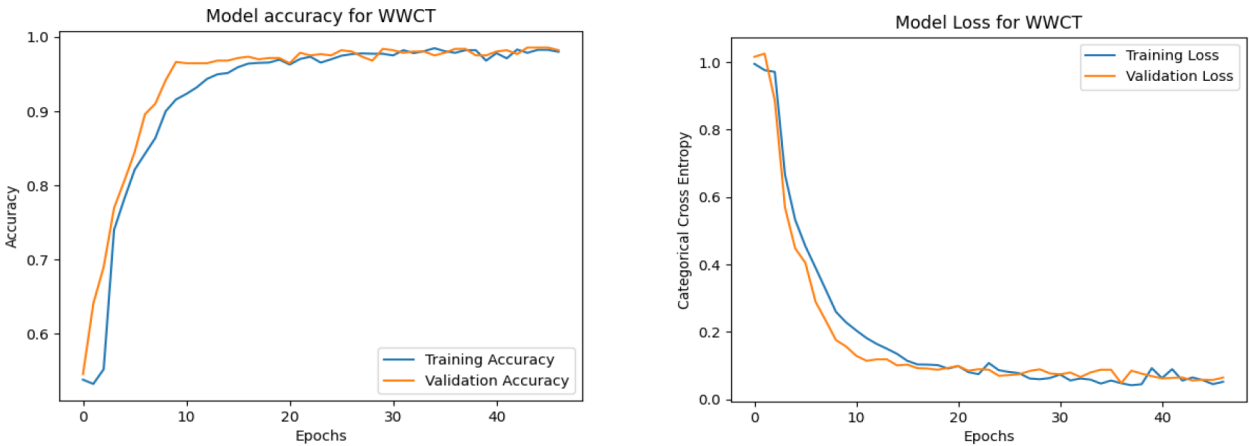


Figure 6—The DCNN history plots for WWCT vector multi-classification.

Fig.7 shows the training history of the DCNN model for WOPR multi-classification.

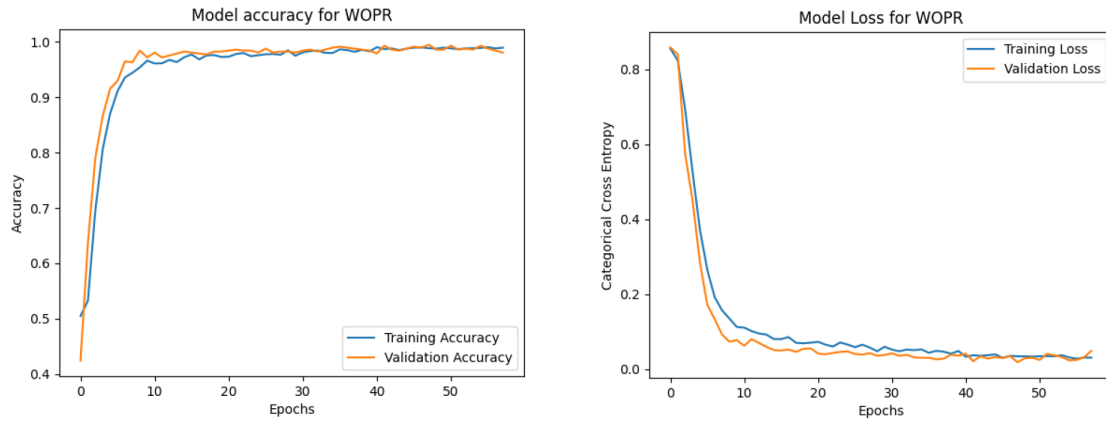


Figure 7—The DCNN history plots for WOPR vector multi-classification.

Fig.8 shows the training history of the DCNN model for WWCT binary classification.

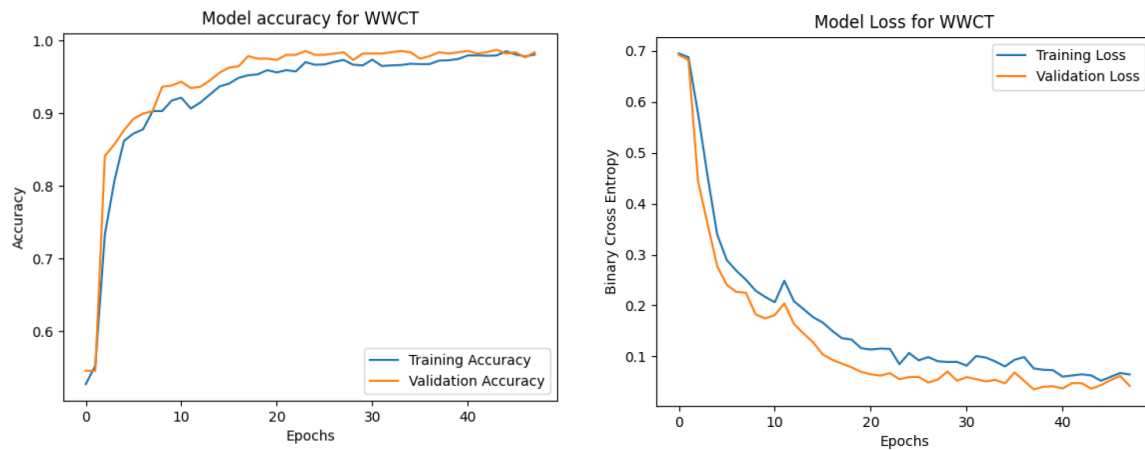


Figure 8—The DCNN history plots for WWCT vector binary classification.

Fig.9 shows the training history of the DCNN model for WOPR binary classification.

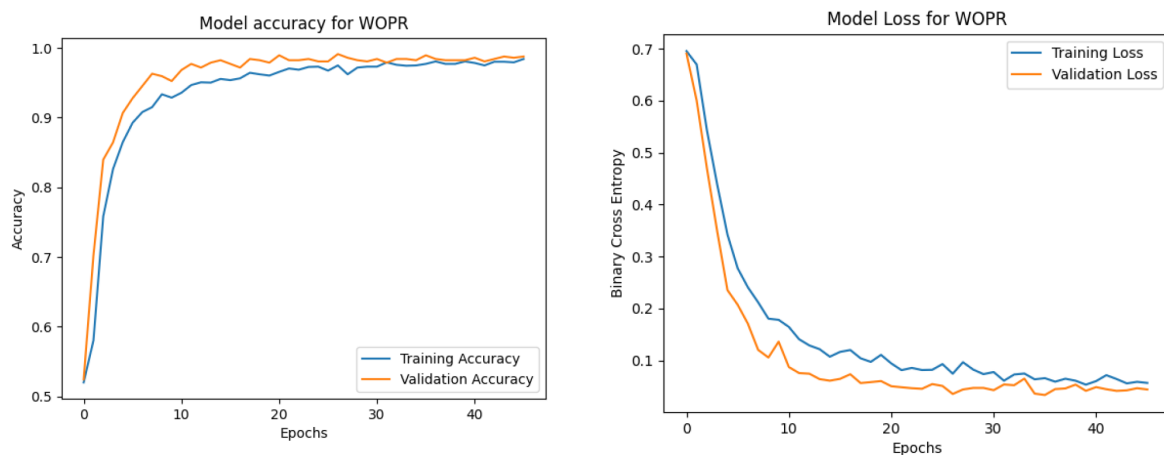


Figure 9—The DCNN history plots for WOPR vector binary classification.

MobileNetV2 Model Evaluation

Table 2 presents the results of the MobileNetV2 model in classifying temporal reservoir data, indicating that the model achieved an accuracy of 95% for the WWCT vector and 93% for the WOPR vector in the

multi-classification scenario. The model achieved 98% for WWCT and 97% for WOPR vector in the binary classification scenario.

Table 2—The results of MobileNetV2 model

Model	Multi classification	Binary classification
MobileNetV2 for WWCT	95%	98%
MobileNetV2 for WOPR	93%	97%

Fig.10 shows the training history of the MobileNetV2 model for WWCT multi-classification.

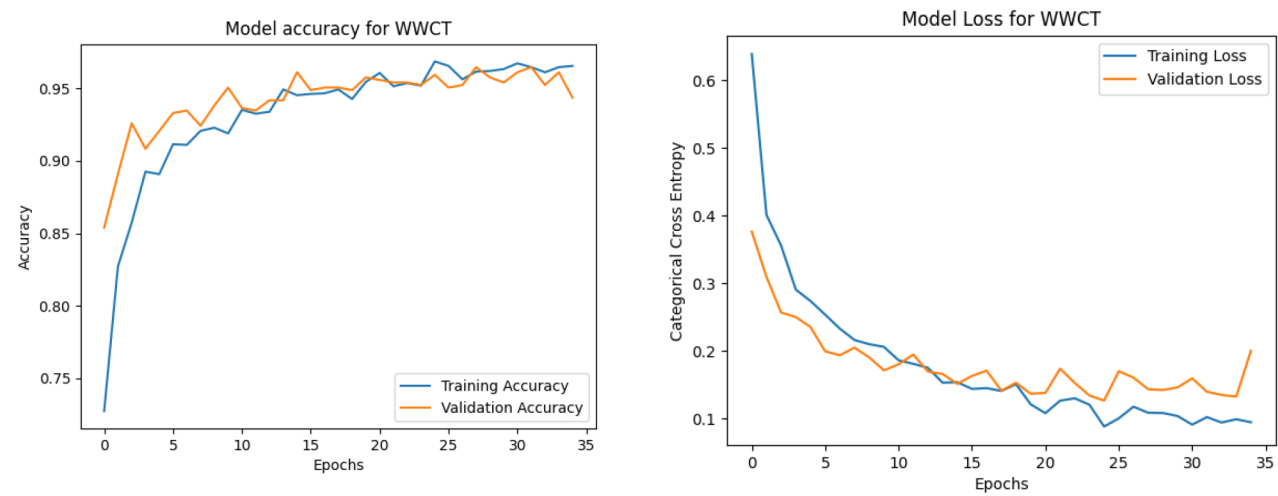


Figure 10—The MobileNetV2 history plots for WWCT vector multi-classification.

Fig.11 shows the training history of the MobileNetV2 model for WOPR multi-classification.

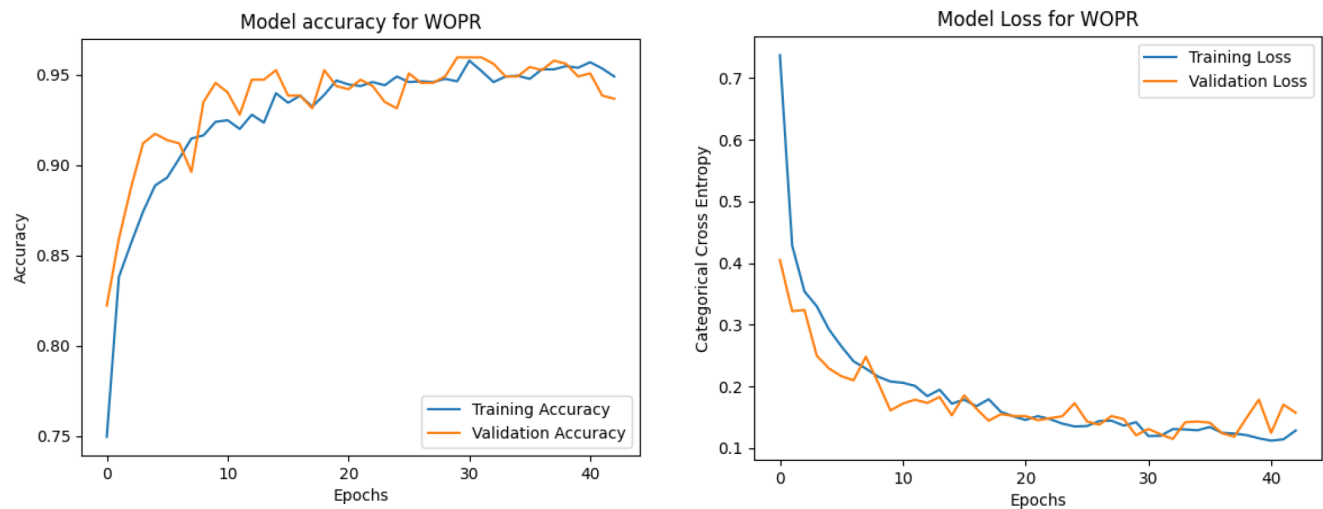


Figure 11—The MobileNetV2 history plots for WOPR vector multi-classification

Fig.12 shows the training history of the MobileNetV2 model for WWCT binary classification.

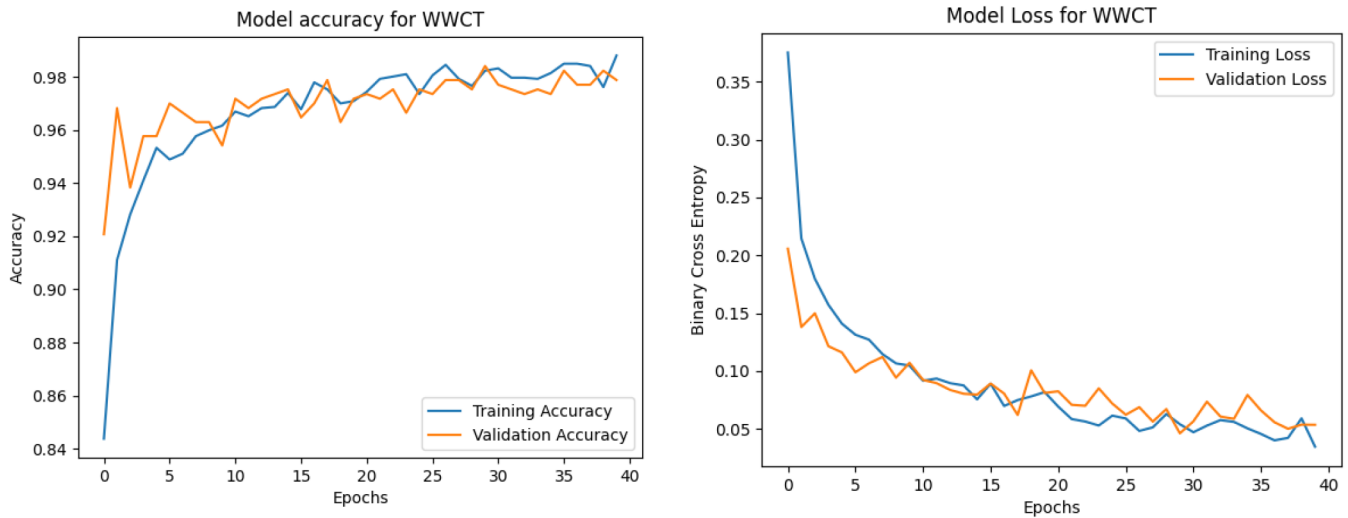


Figure 12—The MobileNetV2 history plots for WWCT vector binary classification

Fig.13 shows the training history of the MobileNetV2 model for WOPR binary classification.

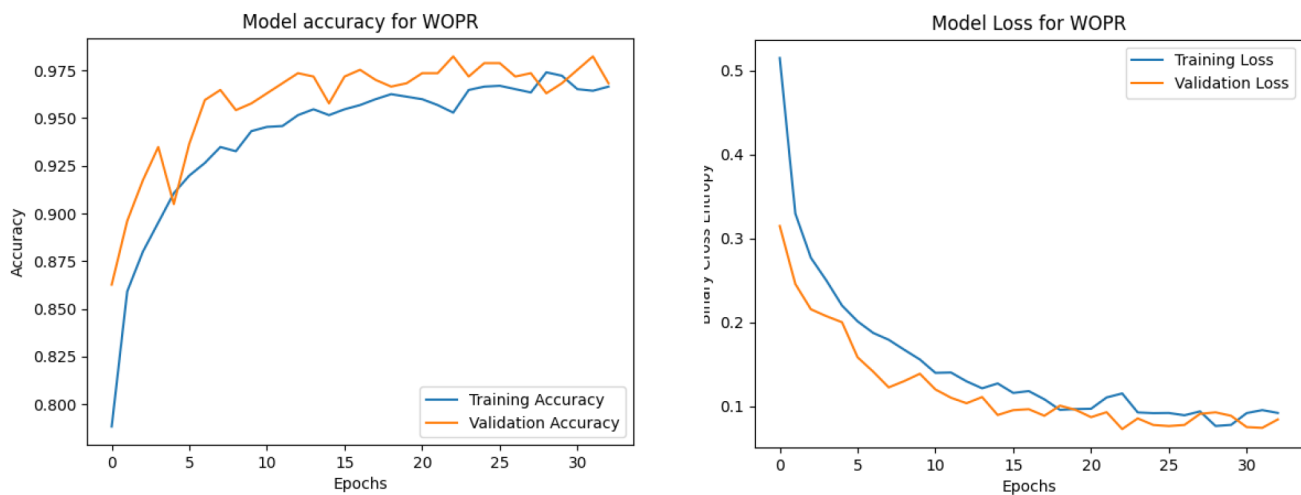


Figure 13—The MobileNetV2 history plots for WOPR vector binary classification

VGG16 Model Evaluation

The results of the VGG19 model in classifying temporal reservoir data are presented in Table 3, which shows that the model achieved an accuracy of 93% for the WWCT vector and 91% for the WOPR vector in the multi-classification scenario, and 97% for both WWCT and WOPR vectors in the binary classification scenario.

Table 3—The results of VGG16 model

Model	Multi classification	Binary classification
VGG16 for WWCT	93%	97%
VGG16 for WOPR	91%	97%

Fig.14 shows the training history of the VGG16 model for WWCT multi-classification.

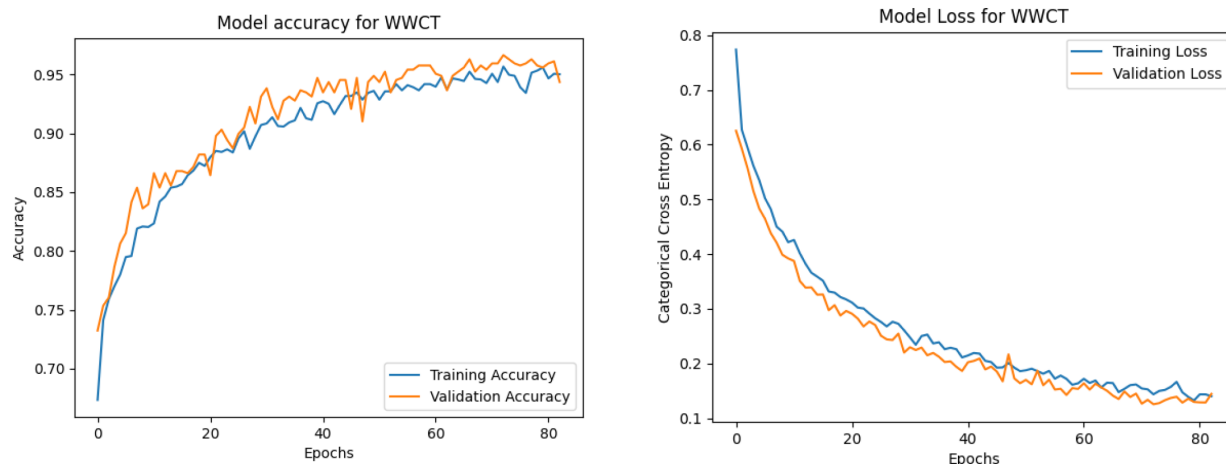


Figure 14—The VGG16 history plots for WWCT vector multi-classification

Fig.15 shows the training history of the VGG16 model for WOPR multi-classification.

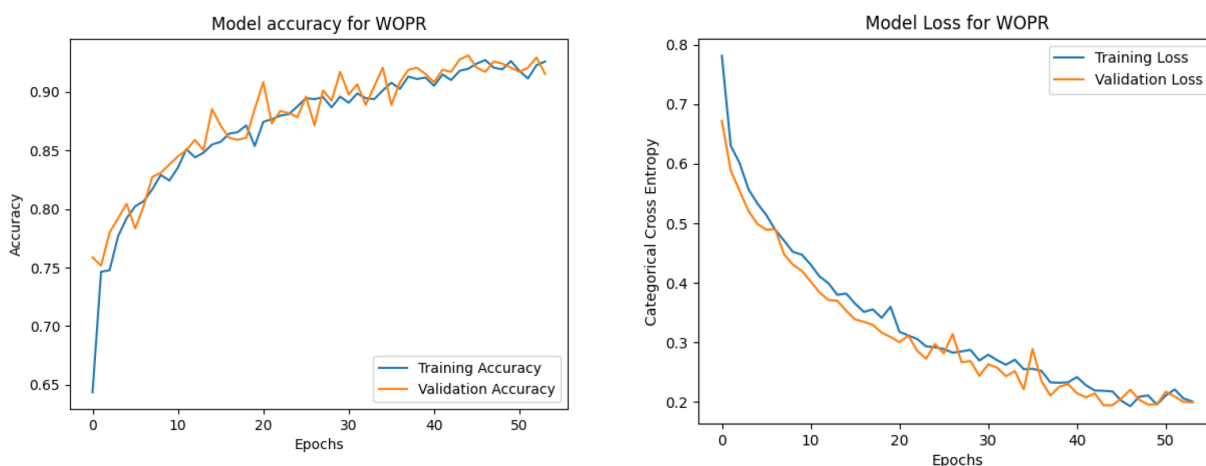


Figure 15—The VGG16 history plots for WOPR vector multi-classification

Fig.16 shows the training history of the VGG16 model for WWCT binary classification.

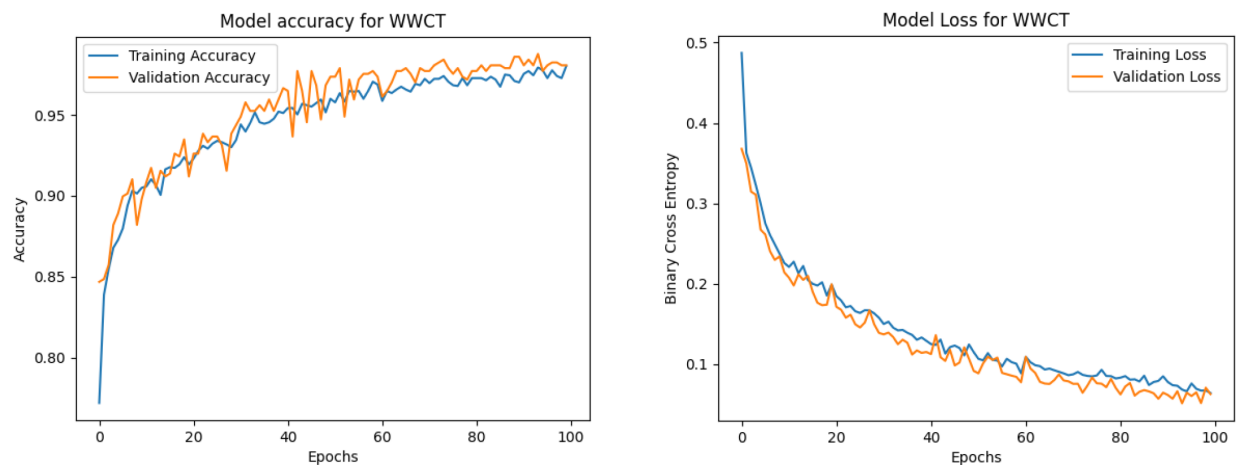


Figure 16—The VGG16 history plots for WWCT vector binary classification.

Fig.17 shows the training history of the VGG16 model for WOPR binary classification.

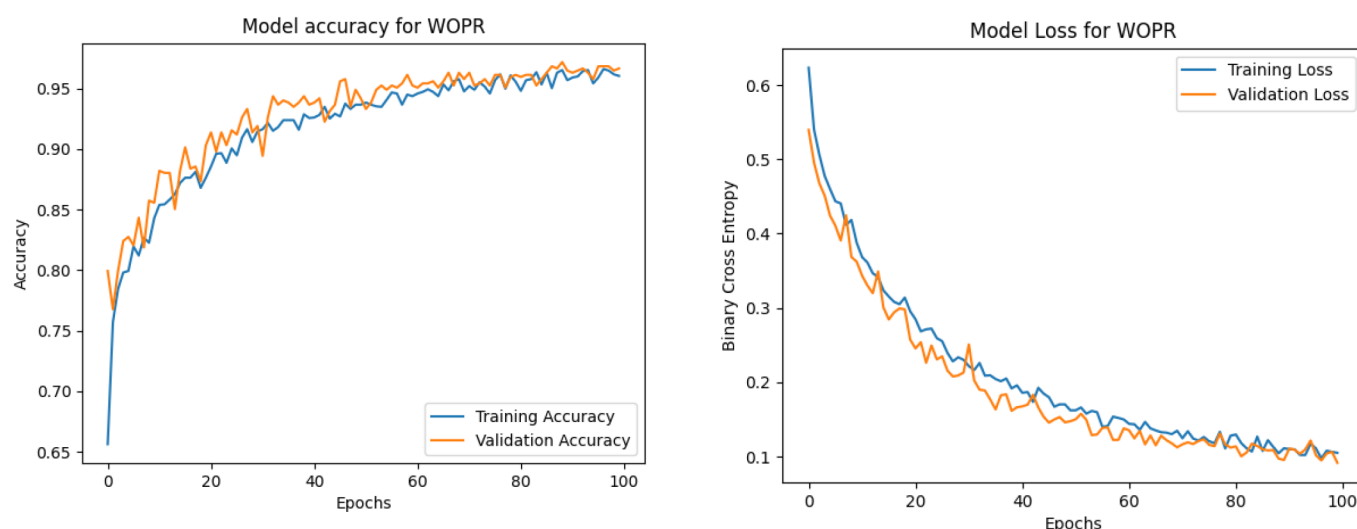


Figure 17—The VGG16 history plots for WOPR vector binary classification.

Conclusion

The Deep Convolutional Neural Network (DCNN) models developed in this study have achieved remarkable results in classifying temporal reservoir data, with an accuracy of 97% for the WWCT vector and 98% for the WOPR vector in the multi-classification scenario, and 98% for both vectors in the binary classification scenario. These results demonstrate the effectiveness of DCNN models in learning complex patterns in temporal reservoir data.

Notably, the DCNN models have significantly reduced the time and effort required for data analysis. Manual labeling of temporal reservoir data can take days, while traditional data analysis and mining methods can take hours to produce results. In contrast, the DCNN models can produce accurate results in a matter of seconds, making them a highly efficient and valuable tool for reservoir engineers.

This has significant implications for the oil and gas industry, where rapid and accurate analysis of temporal reservoir data is critical for optimizing reservoir performance and making informed decisions. By leveraging the power of DCNN models, reservoir engineers can gain valuable insights into reservoir behavior, identify potential issues, and optimize production strategies, all in a fraction of the time required by traditional methods. Overall, the results of this study highlight the potential of DCNN models to revolutionize the field of reservoir engineering and improve the efficiency and accuracy of data analysis.

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